

To study color vision.



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OBJECTIVES:

Student should be able to.

- a) Remember light spectrum & Visual Spectrum.
- b) Explain young Helmholtz and modern color theories.
- c) Interpret Ishihara chart
- d) Perform the procedure.
- e) Certify different color blindness.
- f) Know the genetic involvement in color blindness

- Colors provoke emotional feelings and they are part of everything, from nature and rainbows to man-made creations.
- Colors have a physiological, psychological and social impact on a persons health, wellbeing and status in the world; from the positive stimulating effects of warm colors, to the mental relaxation and soothing effects of cool colors
- Human eye can recognize about 150 different colors in the visible spectrum.
- Discrimination and appreciation of colors depend upon the ability of receptors in retina

Theory

Color vision is a function of cones present in the retina.

Human eye can detect almost all gradations of color, when red, green and blue monochromatic lights are appropriately mixed in different combinations.

Green, yellow, orange and red are colors between wavelength are 525-675 nm. These colors are distinguish one from the other entirely by red and green cones. If either of these two cones are missing one would no longer be able to distinguish between these four colors.

A person with loss of red cone is called Protanope while a color blind person who lacks green cone is called deutanope.

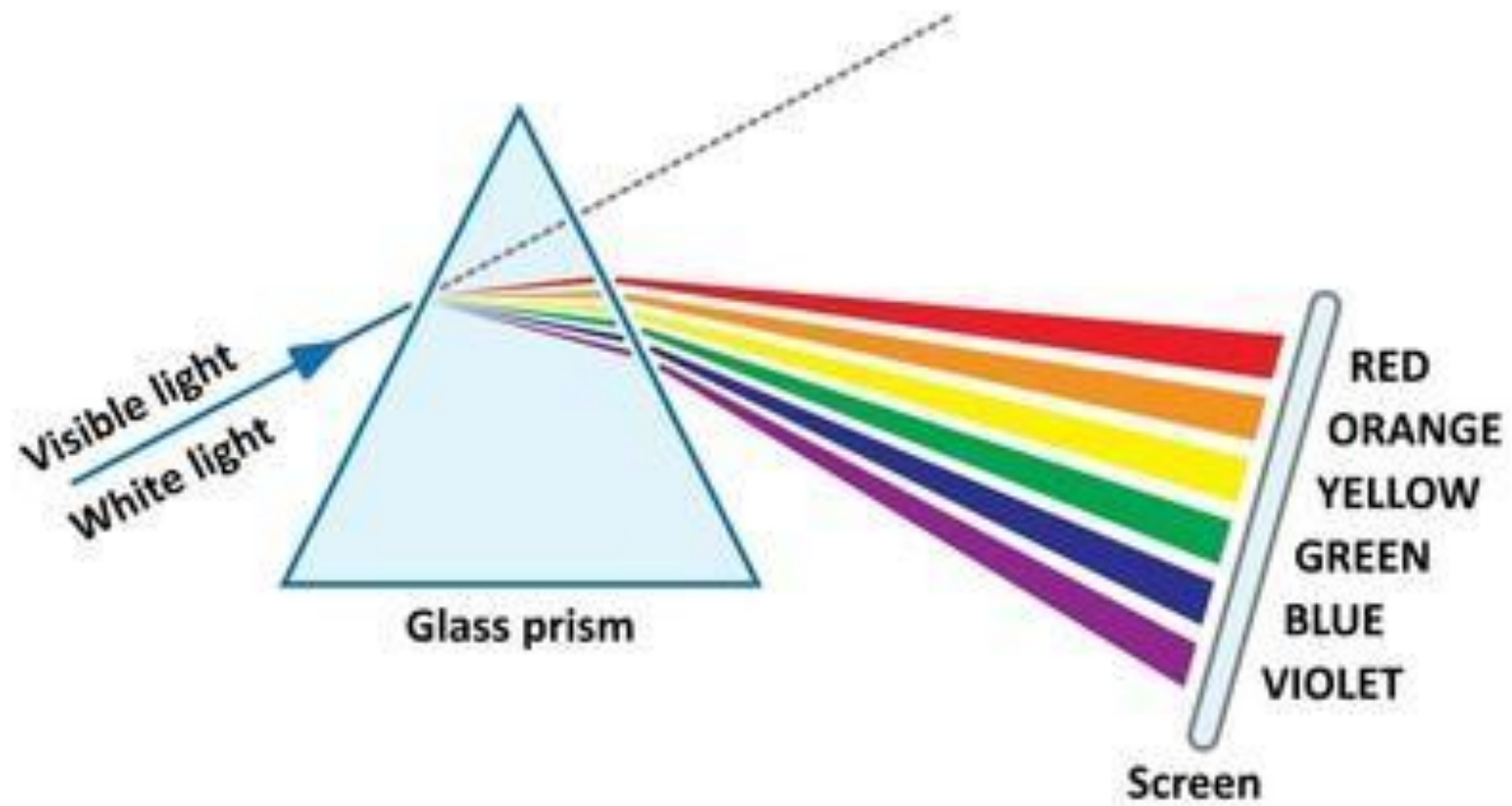
Different colors stimulate cones in different ratios. These ratios are sent to brain which perceives a particular color by a particular ratio.

THEORETICAL EXPLANATION

- The portion of the electromagnetic spectrum with wave lengths between 400 and 700 n.m is the stimulus for sight and is called visual spectrum.
- The electromagnetic spectrum is not colored but in human it stimulates the photo pigments contained in three different types of cones.
- The different signals stimulated by a particular wave initiate a nervous impulse that is further processed by neural elements in the retina and brain so the color is perceived.
- The retina can detect almost all different gradations of colors in the visual spectrum, when red, green, and blue monochromatic lights are appropriately mixed in different combination

VISIBLE SPECTRUM

- **SPECTRAL COLORS** When sunlight or white light is passed through a glass prism, it is separated into different colors. Series of colored light produced by the prism is called the visible spectrum.
- Colors that form the spectrum are called spectral colors.
- Spectral colors are red, orange, yellow, green, blue, indigo and violet).
- In the spectrum, colors occupy the position according to their wavelengths.
- Wavelength is the distance between two identical points in the wave of light energy.
- Accordingly, red has got the maximum wavelength of about 8,000 Å and the violet has got the minimum wavelength of about 3,000 Å.
- Light rays longer than red are called infrared rays. Rays shorter than violet are called ultraviolet rays. But, these two extraordinary types of rays do not evoke the sensation of vision.
- Refraction of spectral colors by the prism also depends on wavelengths. Red is refracted less and violet is refracted more. So, longer the light rays, lesser is the refraction by the prism



THOMAS YOUNG TRICHROMATIC THEORY

According to Thomas Young, retina has three types of cones. Each one possesses its own photosensitive substance. The trichromatic theory of color vision says that human eyes only perceive three colors of light: red, blue, and green.

Each cone gives response to one of the primary colors – red, green and blue.

Different color sensations are produced by the stimulation of various combinations of these three types of cones. For sensation of white light, all the three types of cones are stimulated equal

The wavelengths of these three colors can be combined to create every color on the visible light spectrum.

This theory was first suggested by Thomas Young and later tested by Hermann Von Helmholtz.

YOUNG HELMHOLTZ THEORY

Trichromatic color vision occurs because of stimulation of three types of cones in retina, each of which responds maximally to a different color.

- 1) Red cones are sensitive to Long wave length sensitive cones – L cones – maximally stimulates in the region of 563nm have maximum sensitivity in the region of 563 n.m. also called red cones.
 - 2) Green cones, are stimulated by light rays of middle wave length sensitive cones – M cones – have a maximum sensitivity in the region of 535 n.m. also called green cones.
 - 3) Blue cones Short wave length sensitive cones – S cones – have a maximum sensitivity in the region 420 n.m. also called blue cones.
- Stimulation of these three cones in different ratios results in perception of color by the brain e.g.
 - The message of orange color is sent to brain as 97:42:0 (R:G:B)
 - Yellow color is perceived as 83:83:0 R:G:B)
 - The ratio of blue color is 0:0:97 (R:G:B)
 - Equal stimulation of red, green, and blue gives white color.

MODERN COLOR THEROIES: -

Hyoththesize that there are 3 types of cones.

S Cones – Signal red and blue

M Cones – Signal green and yellow and possibly blue.

L Cones – Signal red and yellow

NEURAL MECHANISM

Color is mediated by ganglion cells that subtract or add input from one type of cone to input from another type. Processing in the ganglion cells and the geniculate nucleus produces three types of neural pathways to the visual cortex.

- 1) A Red – Green pathway that signals difference between L and M cone responses.
- 2) A Blue – Yellow Pathway that signals differences between S cone and sum of L and M cone responses
- 3) A Luminance pathway that signals the sum of L and M cone responses. These pathways project to the blobs and deep portion of layer 4c of visual cortex

APPLIED PHYSIOLOGY

- **COLOR BLINDNESS** Color blindness is the failure to appreciate one or more colors. It is common in 8% of males and only in 0.4% of females, as mostly the color blindness is an inherited sex-linked recessive character.
- In addition to hereditary conditions, color blindness occurs due to acquired conditions also such as ocular diseases or injury or disease of retina. The term 'color blind' does not mean that objects are seen only in black and white.
- Total color blindness is very rare. There are many types and degrees of color blindness.
- Most common type of blindness is red –green ,found in 6 to 8 % male population
- The most appropriate term for color blindness is deficiency of color vision.

Colour blindness is of 3 types.

- 1) Protanope: -** The person with loss of red cones is called a protanope
- 2) Deuteranope:-** The colour blind person who lacks green cones is called a deuteranope
- 3) Tritanope: -** The colour blind person who lacks blue cones

ISHIHARA CHART

The colour vision can be tested by the use of Pseudo isochromatic plates

The Ishihara chart is the most widely used screening tool for diagnosing red-green color vision deficiencies (such as protanopia and deuteranopia). It consists of a series of colored plates (ranging from 10 to 38) made of randomized dots, where individuals with normal vision see a hidden number or line that blends in for those with deficiencies.

Procedure

- Color vision testing should be done in a room well lit by daylight.
- The plates are held at a distance of 75 cm (30 inches) from the subject and tilted so that the plane of paper is at right angle to the line of vision.
- The numbers which are seen from the plates vary within 1-75 and the subject is supposed to give correct answer within 30 seconds.
- If the person is unable to read the numerals, then the charts numbered 26-38 are used and the subject is asked to trace the lines between the lines X's with a brush. Each tracing should be completed within 10 seconds.
- Charts 22,23,24,25, are omitted if the test is designed merely to separate the color deficiency from those with normal color appreciation.
- The sizes and colors of numbers are arranged in such a manner that the person with normal colour vision reads a correct number while a colour blind person reads a different number on the same chart

In Each Case Note The Following.

- 1) Time taken by the subject to read the card.
- 2) Any correction made by the subject patient.
- 3) Any other remarks made by the subject.
- 4) Whether the subject appears to read the number easily or with difficulty

PRECAUTIONS:

- 1) Chart must be appreciated in a room lit adequately by bright light.
- 2) Direct sun light should be avoided, as it can produce discrepancy in result
- 3) Chart must be held at 75 cm distance from the subject so that the plane of the paper is a right angle to the line of vision.
- 4) It is necessary to vary the order of chart if it is suspected that there is deliberate deception on the part of subject.